

Yasir Ali, PhD

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 Zhengzhou, Henan, China

 Research Interests: Secure and Resilient Cyber-Physical Systems (CPSs), Polar Code, UAVs and AI/ML

Research Profile

My research focuses on secure and resilient CPSs, with emphasis on communication security, resilient control, networked control systems, Polar Codes, UAV systems, chaotic encryption, and AI/ML-enabled optimization. My work bridges control theory, physical-layer security, coding theory, and intelligent autonomous systems, with publications in leading journals including IEEE Transactions and Elsevier journals.

Education

- 2018.09 – 2024.06  **Ph.D.** Control Science and Engineering,  [Beijing Institute of Technology, China](#)
- School of Automation
- Thesis: *Enhancing Physical Layer Communications and Security in Optimized CCSs*
- Adviser: Professor Yuanqing Xia, *Fellow, IEEE*
- 2015.09 – 2017.07  **MS.** Control Science and Engineering,  [Beijing Institute of Technology, China](#)
- School of Automation
- Thesis: *A Detection and Mitigation Approach for CCSs against DDoS Attack*
- Adviser: Professor Yuanqing Xia, *Fellow, IEEE*
- 2009.09 – 2014.09  **B.Sc.** Electrical Engineering,  [University of Engineering and Technology, Peshawar](#)
- Department of Electrical (Communication) Engineering
- Thesis: *Ultrasonic Radar System*
- Adviser: Assistant Professor Sahib Khan

Employment History

- 2024.10 – Present  **Postdoctoral Researcher**, [Zhengzhou Research Institute, Beijing Institute of Technology](#)
My postdoctoral research primarily focuses on secure and resilient CPSs, integrating Polar Codes, UAV systems, chaotic systems, and AI/ML techniques to improve communication security, reliability, and system resilience. Several of my research findings have been published in leading international journals.
- 2021.10 – 2024.09  **Visiting Scholar**, [Silesian University of Technology, Gliwice, Poland](#)
Worked as a full-time researcher and instructor at the Faculty of Automatic Control, Electronics and Computer Science. Contributed to undergraduate teaching and conducted advanced research, resulting in high-impact publications in IEEE TII and TNNLS.
- 2014.12 – 2015.08  **Electrical Engineer**, [Power Plant, Saif Textile Mill Limited Gadoon \(Swabi\), KP, Pakistan](#)
Installed and commissioned Caterpillar Gas Gensets (G3520C, G3516A/B), Diesel Genset D3512, and a Waste Heat Recovery Boiler; maintained, operated, and repaired 11kV HT and 440V LT VCBs, OCBs, and other electrical systems.
- 2014.09 – 2014.12  **Internship**, [Rahman Cotton Mill Takht Bhai Pvt. \(Ltd.\), KP, Pakistan](#)
Learned about different types of AC and DC motors and gained practical knowledge in electrical work within the industry.
- 2013.08 – 2013.10  **Internship**, [Malakand Hydro Power Complex \(Malakand-III\), KP, Pakistan](#)
Gained hands-on experience with power plant layout, its equipment and subsystems, operation methodology, and electrical maintenance techniques.
- 2013.06 – 2013.08  **Internship**, [Pakistan Telecommunication Company Limited \(PTCL\)](#)
Gained hands-on experience in operating and maintaining digital exchanges, OMCs/NMCs (EWSD, ZTE), and worked with CDAS, trunk booking, and telecom system quality assurance.

Selected Journal Publications

- J[1] **Yasir Ali**, T. Manzoor, H. Yang, L. You, R. Ma, C. Yan, T. Wang, Y. Xia, Fellow, IEEE, Efficient Multi-Dimensional Pipelined Chaotic Bulk-Codewords-Encryption for Cloud Control Systems, *IEEE Transactions on Cybernetics*. 55, 5014 - 5026 (2025), (SCI, Impact Factor: **10.5**), [doi](#).
- J[2] **Yasir Ali**, Y. Xia, W. Sulek, and T. Manzoor, Chaos Encryption-Based Secure Polar Coding for Network-Oriented Cloud Control System, *IEEE Transactions on Industrial Informatics*, 20, 3935-47 (2023). (SCI, Impact Factor: **9.9**), [doi](#).
- J[3] **Yasir Ali**, T. Manzoor, H. Yang, and Y. Xia, Fellow, IEEE, Artificial Intelligence Driven Channel Coding and Resource Optimization for Wireless Networks, *Computer Science Review*, (Accepted). (SCI, Impact Factor: **12.7**).
- J[4] H. Yang, Y. Ma, H. Xie, **Yasir Ali**, L. Dai, and Y. Xia, Fellow, IEEE, Resilient Output Feedback tube-based MPC for Cyber-Physical Systems Under Hybrid Attacks, *IEEE Transactions on Circuits and Systems I: Regular Papers*, (SCI, Impact Factor: **5.2**), [doi](#).
- J[5] T. Manzoor, H. Pei, Member, IEEE, Z. Sun, Member, IEEE, and **Yasir Ali**, Compound Learning-Based Model Predictive Control Approach for Ducted-Fan Aerial Vehicles, *IEEE Transactions on Neural Networks and Learning Systems*, 36, 9395-9407 (2024). (SCI, Impact Factor: **8.9**), [doi](#).
- J[6] R. Ma, Y. Zhan, C. Wu, Z. Hong, **Yasir Ali**, and X. Yuanqing, Fellow, IEEE, QORA: Neural-Enhanced Interference-Aware Resource Provisioning for Serverless Computing, *IEEE Transactions on Automation Science and Engineering*, 22, 10609 - 10624 (2025), (SCI, Impact Factor: **6.4**), [doi](#).
- J[7] **Yasir Ali**, Y. Xia, Fellow, IEEE, T. Manzoor, S. Ali, M. Abouhawwash, S. S. Askar, A. K. Kumar, Member, IEEE, and R. Ma. Efficient Hardware Realization of SC Polar Decoders Using Compound Pipelined Processing Units and Auxiliary Registers, *IEEE Access*, 12, 23808-23826 (2024), (SCI, Impact Factor: **3.6**), [doi](#).
- J[8] **Yasir Ali**, Y. Xia, L. Ma, and A. Hammad, Secure Design For Cloud Control System Against Distributed Denial of Service Attack, *Control Theory and Technology*, 16, 14-24 (2018). (SCI, Impact Factor: **1.5**), [doi](#).
- J[9] **Yasir Ali**, and Y. Xia, Reliability Weight-based Extended Puncturing Mechanism for Polar Codes Using Translation Shift, *Transactions on Emerging Telecommunications Technologies*, 35, e4923, (2024), (SCI, Impact Factor: **2.5**), [doi](#).
- J[10] A. K. Kumar, **Yasir Ali**, R. R. Kumar, M. H. Assaf, and S. Ilyas, Artificial Intelligent and Internet of Things Framework for Sustainable Hazardous Waste Management in Hospitals, *Waste Management*, 203, 114816 (2025), (SCI, Impact Factor: **7.1**), [doi](#).
- J[11] T. Manzoor, Y. Xia, **Yasir Ali**, and K Hussain, Flight Control Techniques and Classification of Ducted Fan Aerial Vehicles, *Control Theory and Application*. 39 (2), 201-221, (EI), [doi](#).
- J[12] M. Mohasan, M. Song, S. M. A. Naqvi, S. Zhou, M. Hamid, and **Yasir Ali**, Numerical study of double droplet impacts on heated films: influence of film thickness and temperature on crown and jet dynamics, *Meccanica*. Vol. 61, Article No. 48, 2026, (SCI/SCIE, Impact Factor: **2.1**).

Selected Conference Publications

- C[1] **Yasir Ali**, Z. Shen, F. Zhu, G. Xiong, S. Chen, Y. Xia, F. and Y. Wang, Solutions Verification for Cloud-Based Networked Control System using Karush-Kuhn-Tucker Conditions, In Proceedings of Chinese Automation Congress (CAC), November 30-December 02, 2018, Xi'an, China, [doi](#).
- C[2] **Yasir Ali**, and W. Sulek, Measurement Analysis of Error Rate for Belief's Clipping and Quantization in Polar Codes, Radiocommunication and ICT Conference Krakow, September 19-21, 2023, Krakow, Poland, [doi](#).
- C[3] **Yasir Ali**, Y. Xia, and L. Ma, A Stabilizing Architecture for Cloud Control System against DDoS Attack, In Proceedings of Chinese Automation Conference (CAC), October 20-22, 2017, Jinan, Shandong, China, [doi](#).
- C[4] R. Ma, Y. Zhan, T. Yan, Y. Xia, and **Yasir Ali**, Interless: Interference-Aware Deep Resource Prediction for Serverless Computing, In Proceedings of 36th Chinese Control and Decision Conference (CCDC), May 25-27, 2024, Xi'an, China, [doi](#).
- C[5] L. Ma, Y. Xia, **Yasir Ali**, and Y. Zhan, Engineering problems in initial phase of cloud control system, In Proceedings of 36th Chinese Control Conference (CCC), July 26-28, 2017, Dalian, China, [doi](#).
- C[6] H. Ahmad, X. Yao, M. Muddassir, J. Chiragh and **Yasir Ali**, Humanoid robots object grasping and manipulation learning by demonstration, In Proceedings of 3rd International Conference on Control, Automation and Robotics (ICCAR), April 24-26, 2017, Nagoya, Japan, [doi](#).

Ongoing Work

- SJ[1] **Yasir Ali**, T. Manzoor, L. Dai, Y. Xia, Authenticated ECDH-Driven Chaotic Packet Protection for Resilient Cloud Control Systems, *IEEE Transactions on Cybernetics*. (Under review), (SCI, Impact Factor: **10.5**).
- SJ[2] **Yasir Ali**, Reliability-Aware Adaptive CRC-Aided Polar Decoding With Configurable Effort Control, *IEEE Transactions on Communications*. (Under review), (SCI, Impact Factor: **8.3**).

- SJ[3] **Yasir Ali**, T. Manzoor, M. M, Y. Xia, Cyber-Physical Safety Capsule for Cloud-Controlled UAVs Under Availability and Integrity Attacks, *ACM Transactions on Cyber-Physical Systems*. (Under review), (ESCI, Impact Factor: **2.9**).
- SJ[4] **Yasir Ali**, T. Manzoor, H. Yang, A. Ali, Y. Xia, Towards Efficient and Secure Cloud Control Systems: Advances, Challenges, and Future Directions, *Annual Reviews in Control (Elsevier)*. (Under review), (SCI, Impact Factor: **10.7**).
- SJ[5] T. Manzoor, **Yasir Ali**, Y. Xia, L. You, Y. Wang, Physics-informed sparse identification-based tube model predictive control for aerial vehicles, *Engineering Applications of Artificial Intelligence*. (Under review), (SCI, Impact Factor: **8.0**). Preprint available at *arXiv:2605.23354*, DOI: 10.48550/arXiv.2605.23354.
- SJ[6] C. Yan, W. Zhang, L. Yan, B. Dong, Q. Wang, **Yasir Ali**, Yuanqing Xia, IEEE Fellow, Event-Sensory-Feedback Approach to Distributed Attack Isolation in Resilient Output Consensus, *IEEE Transactions on Cybernetics*. (Under review), (SCI, Impact Factor: **10.5**).

Research Grants & Projects

- 2026.01 – 2026.12  **National Natural Science Foundation of China (NSFC)**, Principal Investigator (PI), Grant No. **W2533176, RMB 200,000**. One-year project under the category of *Research Fund for International Young Scientists*.
- 2026.01 – 2026.12  **Beijing Natural Science Foundation (BJNSF)**, PI, Grant No. **IS25064, RMB 200,000**. One-year project under the category of *International Scholars*.
- 2025.12 – 2026.09  **High-end Foreign Experts Project of Henan Province**, selected as a high-level foreign expert for a fully funded 10-month research program, Grant No. **HNGD2026007**, with total project funding of **RMB 150,000**.
- 2026.01 – 2026.12  **Henan Postdoctoral Foundation Research Fund**, PI, Grant No. **HN2026058, RMB 100,000**. One-year project under the category of *Henan Postdoctoral Research Funding Scheme*.

Honors & Awards

- 2025.12 – 2028.09  **Senior Professional Title — Associate Researcher**, awarded the Natural Science Research Series Senior Professional Title of *Associate Researcher* in Henan Province, China.
- 2018.09 – 2019.07  **Best Paper Award**, awarded by *Control Theory and Technology* for an outstanding research article published in the 2018–2019 volume.
- 2015.09 – 2023.07  **Chinese Government Scholarship**, awarded for Master's and Doctoral studies in Beijing; a competitive scholarship supporting full tuition and living expenses.
- 2011.03 – 2012.04  **Academic Excellence Award**, awarded by the Khyber Pakhtunkhwa Provincial Government for achieving a cumulative score of 80% across Bachelor's examinations.

Hardware & Computer Skills

Programming: C, C++, Python. **Numerical Analysis and Optimization Tools:** MATLAB (solvers: YALMIP, CasADi, OSQP), Octave. **Hardware:** FPGA design with VHDL and Verilog. **Tools for Instrumentation, Control, Data Acquisition:** LabVIEW, Simulink. **Drafting & Productivity:** $\text{\LaTeX}$, Microsoft Office, WPS, Inkscape. **Computer Aided Design Tool:** FreeCAD. **Operating Systems:** Windows, Ubuntu, MAC.

Language Competencies

English (Excellent) | Urdu (Excellent) | Chinese (Fair Level) | Polish (Basic Level)

References

Professor Yuanqing Xia
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